APPLIED MATERIALS • GENAI INFRASTRUCTURE TEAM

Software Student Take-Home Exercise

Full Stack & Agentic AI Infrastructure • June 2026

Thank you for your interest in joining the **GenAI Infrastructure team** at Applied Materials. This exercise gives you the opportunity to demonstrate your skills in a practical, open-ended way — the same kind of work you would be doing with us from day one.

Please **choose ONE of the two exercises below** and implement it. Both require a working GitHub repository and a deployed, accessible application. We value clean code, good documentation, and thoughtful design decisions over feature completeness.

Time frame: You have **3–7 days** to submit from when you receive this document. Submitting early allows us to schedule your technical interview sooner.

— CHOOSE ONE EXERCISE —

OPTION A

Full-Stack Dataset Explorer with AI Insights

Build a web application where a user uploads a CSV dataset, explores its contents, and asks natural-language questions — answered by an LLM of your choice.

BACKEND — PYTHON / FASTAPI

- `POST /upload` — accept a CSV, parse it into memory or SQLite
- `GET /rows` — return rows with basic filtering or pagination
- `POST /ask` — accept a free-text question, query an LLM with relevant context, return a structured answer

FRONTEND — REACT OR ANY MODERN FRAMEWORK

- File upload UI
- Filterable / searchable table view
- "Ask a question" input with LLM response panel

WHAT WE EVALUATE

- API design clarity (endpoint naming, request/response shapes)
- React state management and user experience
- How you structure the LLM prompt and handle its response
- Code quality and README clarity

OPTION B

Telegram Bot with AI Capabilities

Build a Telegram bot that acts as a personal AI assistant, exposing three core capabilities via commands or natural conversation.

REQUIRED FEATURES

- **Chat with LLM** — multi-turn conversation with a language model of your choice (OpenAI, Claude, Gemini, etc.)
- **Shallow web search** — quick retrieval of top results for a query
- **Deep web search** — follow links, extract content, and synthesize a complete answer
- **Image generation** — generate an image from a text prompt and send it back in the chat

WHAT WE EVALUATE

- How you handle multi-turn conversation state
- Quality of the deep search pipeline (synthesis vs. link dump)
- Code structure and separation of concerns
- Reliability of the deployed bot

General Requirements

These requirements apply to **both** exercises.

LANGUAGE

Python backend preferred; any modern frontend framework

LLM & IMAGE APIS

Any provider you're comfortable with — OpenAI, Claude, Gemini, DALL-E, Flux, etc.

DEPLOYMENT

Any free-tier platform: Railway, Render, Fly.io, Vercel, etc.

REPOSITORY

Public GitHub repo with a working README

Security: Never commit API keys or credentials to the repository. Use environment variables for all secrets, and document clearly in your README which variables need to be configured.

Deliverables

1. **Public GitHub repository** with a README covering: how to run locally, how to deploy, which environment variables to set, and a brief architecture overview.
2. **Live deployed application** accessible via a public URL (Option A) or a working Telegram bot link (Option B).
3. **"What I'd do next"** section in the README — 2 to 5 bullet points describing improvements or features you would add given more time.

We are not looking for a perfect, production-grade system. We want to see how you think: how you structure a solution, how you handle edge cases, and how you communicate your design decisions. A clean, well-documented partial implementation is more valuable to us than a messy but feature-complete one.

How to Submit

1. Make sure your GitHub repository is **public** and your deployed application or bot is live and accessible.
2. Send an email with the subject line **[Take-Home Submission] Your Name** to both contacts listed below.
3. Include in the email: your **GitHub repository URL**, your **deployed app or bot URL**, and any notes about your implementation or design choices you'd like us to know.

Questions & Contact

If you have any questions about the exercise, need clarification on requirements, or run into a technical issue — please reach out. We aim to respond within one business day.

Don't hesitate to ask. We'd rather you ask a question than spend time going in the wrong direction.

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